Balanced forces can mean steady motion

LEARNING TARGET · *R6* **I can explain steady motion with Newton's first law.**

A · KEY VOCABULARY

Inertial frame	An unforced object keeps its velocity.
Net force	Vector sum of external forces on a system.
Inertia	More mass resists velocity change more.
Equilibrium	Net force zero; v constant, not always zero.

B · KEY EQUATIONS + UNITS

$$\Sigma F = 0 \Rightarrow a = 0$$

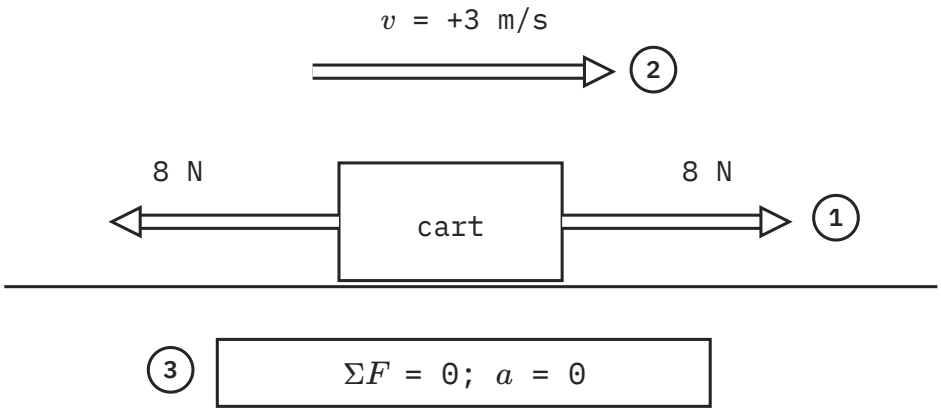
$$x_f = x_0 + v\Delta t$$

Σ sum on one object

v stays constant

C · COLOR IT · BALANCED AND MOVING

Right is +.



☐ **1 · ORANGE** Opposing forces

☐ **2 · PURPLE** Rightward velocity

☐ **3 · YELLOW** Net force zero

D · WORKED EXAMPLE

A cart is at **−2 m**, moving at **+3 m/s**. Its external forces balance for the next **4 s**. **Find its final position.**

GIVEN	EQUATION	WORK	ANSWER
x_0 = **−2 m** v_i = **+3 m/s** ΣF = 0; t = **4 s**	First law: constant v. $x_f = x_0 + vt$.	a = **0 m/s²** x_f = −2 + 3(4)	**+10 m** Net force zero; motion continues.

E · THE 2026-XJ THREAD

DAY 9 OF 22 · NEW TOOL: INERTIA

XJ needs no forward force simply to keep moving. Gravity can still change its velocity, so treating a short interval as steady motion is an approximation to be tested.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else